Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The locus of the point from which mutually perpendicular tangents can be drawn to the circle $x^2 + y^2 = 36$ is

(A) $x^2 + y^2 = 42$

(B) $x^2 + y^2 = 48$

(C) $x^2 + y^2 = 60$

(D) $x^2 + y^2 = 72$

Q.2

The locus of the middle points of the chords of the circle $x^2 + y^2 = 4a^2$ which subtend a right angle at the centre of the circle is

(A) $x + y = 2a$

(B) $x^2 + y^2 = a^2$

(C) $x^2 + y^2 = 2a^2$

(D) $x^2 + y^2 = x + y$

Q.3

A variable chord of the circle $x^2 + y^2 - 2ax = 0$ is drawn through the origin. Locus of the centre of the circle drawn on this chord as diameter is

(A) $x^2 + y^2 + ax = 0$

(B) $x^2 + y^2 + ay = 0$

(C) $x^2 + y^2 - ax = 0$

(D) $x^2 + y^2 - ay = 0$

Q.4

If a circle passes through the point $(3, 4)$ and cuts the circle $x^2 + y^2 = a^2$ orthogonally, the equation of the locus of its centre is

(A) $3x + 4y - a^2 = 0$

(B) $6x + 8y = a^2 + 25$

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(C) $6x + 8y + a^2 + 25 = 0$

(D) $3x + 4y = a^2 + 25$

Q.5

The centres of a set of circles, each of radius 3, lie on the circle $x^2 + y^2 = 25$. The locus of any point in the set is

(A) $4 \leq x^2 + y^2 \leq 64$

(B) $x^2 + y^2 \leq 25$

(C) $x^2 + y^2 \geq 25$

(D) $3 \leq x^2 + y^2 \leq 9$

Q.6

A circle touches the x-axis and also touches the circle with centre at $(0, 3)$ and radius 2. The locus of the centre of the circle is

(A) a hyperbola

(B) a parabola

(C) an ellipse

(D) a circle

Q.7

The locus of the mid-point of the chord of contact of tangents drawn from points lying on the straight line $4x - 5y = 20$ to the circle $x^2 + y^2 = 9$ is

(A) $20\left(x^2 + y^2\right) - 36x + 45y = 0$

(B) $20\left(x^2 + y^2\right) + 36x - 45y = 0$

(C) $36\left(x^2 + y^2\right) - 20x + 45y = 0$

(D) $36\left(x^2 + y^2\right) + 20x - 45y = 0$

Q.8

Two rods of lengths a and b slide along the x-axis and y-axis respectively in such a manner that their ends are concyclic. The locus of the centre of the circle passing through the end points is

(A) $4\left(x^2 + y^2\right) = a^2 + b^2$

(B) $x^2 + y^2 = a^2 + b^2$

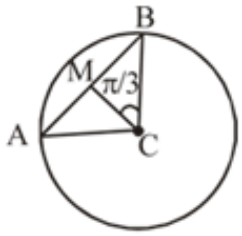
(C) $4\left(x^2 - y^2\right) = a^2 - b^2$

(D) $x^2 - y^2 = a^2 - b^2$

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Q.9

Let C be the circle with centre $(0,0)$ and radius 3 units. The equation of the locus of the mid points of the chords of the circle C that subtend an angle of $\frac{2\pi}{3}$ at its centre is



(A) $x^2+y^2=\frac{9}{4}$

(B) $x^2+y^2=\frac{3}{2}$

(C) $x^2+y^2=1$

(D) $x^2+y^2=\frac{27}{4}$

Q.10

If a circle passes through the point (a,b) and cuts the circle $x^2+y^2=p^2$ orthogonally, then the equation of the locus of its centre is

(A) $x^2+y^2-2ax-3by+\left(a^2-b^2-p^2\right)=0$

(B) $2ax+2by-\left(a^2+b^2+p^2\right)=0$

(C) $x^2+y^2-3ax-4by+\left(a^2+b^2-p^2\right)=0$

(D) $2ax+2by-\left(a^2-b^2+p^2\right)=0$

Q.11

If a line segment $\mathrm{AM}=$ a moves in the plane XOY remaining parallel to OX so that the left end point A slides along the circle $x^2+y^2=a^2$, the locus of M is

(A) $x^2+y^2=4a^2$

(B) $x^2+y^2=2ax$

(C) $x^2+y^2=2$ ay

(D) $x^2+y^2-2ax-2ay=0$

Q.12

A variable circle passes through the fixed point $A(p,q)$ and touches x -axis. The locus of the other end of the diameter through A is

(A) $(x-p)^2=4qy$

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(B) $(y-q)^2=4px$

(C) $(y-p)^2=4qx$

(D) $(x-q)^2=4py$

Q.13

If equation of a circle is given by r^2 - $8r\sqrt{3}\cos\theta+\sin\theta+15=0$ (where $x=r\cos\theta,\quad y=r\sin\theta$), then the radius of the circle is :

(A) 8

(B) 7

(C) 6

(D) 5

Q.14

If the pair of tangents are drawn from $O0,\quad 0$ to the circle x^2+y^2 - $6x$ - $8y=$ - 21 meets the circle in A and B, then length of BA is

(A) 11

(B) $\frac{4}{5}\sqrt{21}$

(C) $\sqrt{\frac{17}{3}}$

(D) $\frac{\sqrt{21}}{5}$

Q.15

The line joining $(5,0)$ to $((10\cos\theta,10\sin\theta)$ is
divided internally in the ratio $2:3$ at P. If θ varies, then the locus of P is

(A) a pair of straight lines

(B) a circle

(C) a straight line

(D) None of these

Answers & Solutions

Q.1 Answer:

$x^2+y^2=72$

Solution:

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$y = mx \pm 6\sqrt{1+m^2}$ is the equation of any tangent to the circle. If it passes through (h, k), then
$k = mh \pm 6\sqrt{1+m^2}$
$\Rightarrow \quad (k-mh)^2 = 36\left(1+m^2\right)$
$\Rightarrow \quad \left(36-h^2\right)m^2 + 2mhk + \left(36-k^2\right) = 0$
which gives two values of m say m_1 and m_2, slopes of two tangents passing thought (h, k). These tangents are perpendicular
If $\quad m_1 m_2 = -1 \Rightarrow \frac{36-k^2}{36-h^2} = -1$
$\Rightarrow h^2 + k^2 = 72 \quad \Rightarrow$ Locus of (h, k) is $x^2 + y^2 = 72$.

Q.2 Answer:

$x^2 + y^2 = 2a^2$

Solution:

Let $M(h, k)$ be the middle point of the chord AB of the given circle $x^2 + y^2 = 4a^2$, with centre at $O(0, 0)$ and radius equal to $2a$. Then OM is perpendicular to AB Since AOB is a right angled triangle

$$\begin{aligned} 4(AM)^2 &= (AB)^2 = (OA)^2 + (OB)^2 \\ &= (2a)^2 + (2a)^2 = 8a^2 \\ \Rightarrow AM &= \sqrt{2}a \end{aligned}$$

Also $(OA)^2 = (OM)^2 + (AM)^2$
$\Rightarrow (2a)^2 = h^2 + k^2 + 2a^2$
$\Rightarrow h^2 + k^2 = 2a^2$
$\Rightarrow$ Locus of (h, k) is $x^2 + y^2 = 2a^2$.

Q.3 Answer:

$x^2 + y^2 - ax = 0$

Solution:

Let (h, k) be the centre of the required circle then (h, k) is the middle point of the chord say OA of the given circle $x^2 + y^2 - 2ax = 0, O$ being the origin. $\Rightarrow \quad A(2h, 2k)$ lies on the circle
$\Rightarrow h^2 + k^2 - ah = 0$
$\Rightarrow \quad$ Locus of (h, k) is $x^2 + y^2 - ax = 0$.

Q.4 Answer:

$6x + 8y = a^2 + 25$

Solution:

Let the equation of the required circle be
$x^2 + y^2 + 2gx + 2fy + c = 0$ ...(i)
Since it passes through $(3, 4)$

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$$9 + 16 + 6g + 8f + c = 0$$

$$\Rightarrow \quad 6g + 8f + c = -25 \quad \text{(ii)}$$

As (i) cuts the circle $x^2 + y^2 = a^2$ orthogonally

$$2g \times 0 + 2f \times 0 = c - a^2$$

$$\Rightarrow \quad c = a^2$$

So from (ii) we get

$$6g + 8f + a^2 + 25 = 0$$

Hence locus of the centre $(-g, -f)$ is

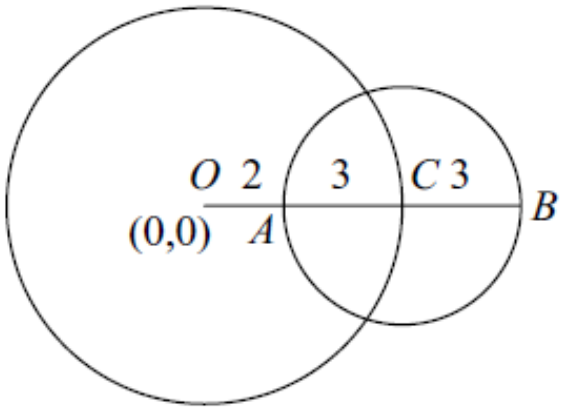
$$6x + 8y - \left(a^2 + 25\right) = 0$$

Q.5 Answer:

$4 \leq x^2 + y^2 \leq 64$

Solution:

Distance of any point (x, y) on a circle with centre C lying on the circle $x^2 + y^2 = 25$ and radius 3 from the origin lies between $OA(= 2)$ and $OB(= 8)$ Hence $4 \leq x^2 + y^2 \leq 64$.



Q.6 Answer:

a parabola

Solution:

Let (h, k) be the centre of the circle. Since it touches x-axis, its radius is k. As it also touches the circle with centre $(0, 3)$ and radius 2 (externally as x-axis lies outside the given circle) distance between the centres is equals to the sum of their radii.

So $h^2 + (k - 3)^2 = (k + 2)^2$

$\Rightarrow h^2 = 5(2k - 1)$

Locus of the centre (h, k) is

$x^2 = 5(2y - 1)$

which is a parabola.

Q.7 Answer:

$20\left(x^2 + y^2\right) - 36x + 45y = 0$

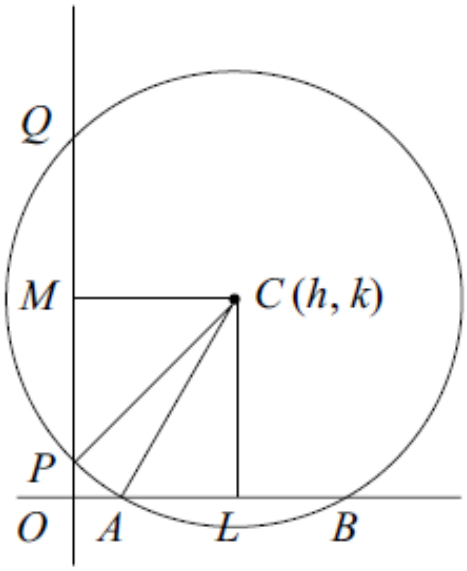
Solution:

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Equation of the chord contact in terms of the mid-point (α, β) is $x\alpha + y\beta = \alpha^2 + \beta^2$
Let $P(5t, 4t-4)$ be a point on $4x - 5y = 20$ Equation of the chord of contact of P is
$x(5t) + y(4t-4) = 9$
comparing the equations we get
$\frac{5t}{\alpha} = \frac{4t-4}{\beta} = \frac{9}{\alpha^2+\beta^2} = \frac{4(5t)-5(4t-4)}{4\alpha-5\beta}$
$\Rightarrow \quad 20\left(\alpha^2+\beta^2\right) - 36\alpha + 45\beta = 0$
Locus of (α, β) is $20\left(x^2+y^2\right) - 36x + 45y = 0$

Q.8 Answer:
$4\left(x^2 - y^2\right) = a^2 - b^2$
Solution:

Let $C(h, k)$ be the centre of the circle passing through the end points of the rod AB and PQ of lengths a and b respectively, CL and CM be perpendiculars from C on AB and PQ respectively. (Fig.)



then $AL = (1/2)AB = a/2$
$PM = (1/2)PQ = b/2$
and $CA = CP$ (radii of the same circle) $\Rightarrow \quad k^2 + \frac{a^2}{4} = h^2 + \frac{b^2}{4}$
$\Rightarrow \quad 4\left(h^2 - k^2\right) = a^2 - b^2$
so that locus of (h, k) is $4\left(x^2 - y^2\right) = a^2 - b^2$

Q.9 Answer:
$x^2 + y^2 = \frac{9}{4}$
Solution:

Let $M(h, k)$ be mid point of the chord AB. In ΔAMC, $MC = BC \cdot \cos \pi/3 = 3 \times 1/2$
$= 3/2$

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$\Rightarrow h^2 + k^2 = 9/4$

$\Rightarrow x^2 + y^2 = 9/4$ **is the locus of the mid point of chords.**

Q.10 Answer:

$2ax + 2by - \left(a^2 + b^2 + p^2\right) = 0$

Solution:

Let the centre be (h, k) **and suppose the equation of the circle is** $x^2 + y^2 - 2hx - 2ky + c = 0$. **The orthogonality condition gives** $c = p^2$. **Substituting this value of** c **and putting** $x = a, y = b$ **in the equation of the circle gives the locus** $2ax + 2by - \left(a^2 + b^2 + p^2\right) = 0$.

Q.11 Answer:

$x^2 + y^2 = 2ax$

Solution:

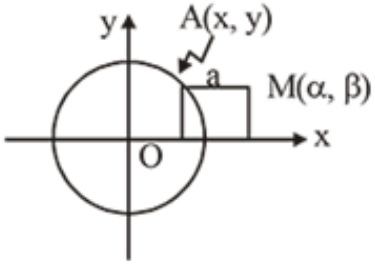
Let the coordinates of A **be** (x, y) **and** Mb (α, β)

Since AM is parallel to OX, $\alpha = x + a$ **and** $\beta = y \Rightarrow x = \alpha - a$ **and** $y = \beta$

As $A(x, y)$ **lies on the circle** $x^2 + y^2 = a^2$ **we have** $(\alpha - a)^2 + \beta^2 = a^2$

$\alpha^2 - 2\mathrm{a}\alpha + \beta^2 = 0$

$\Rightarrow$ **locus of** $M(\alpha, \beta)$ **is** $x^2 + y^2 = 2ax$



Q.12 Answer:

$(x - p)^2 = 4qy$

Solution:

Let the other end of the diameter be (h, k). **The equation of the circle touching the** x **-axis is** $x^2 + y^2 + 2gx + 2fy + g^2 = 0$. **The center** $(-g, -f)$ **satisfies** $p + h = -2g, q + k = -2f$. **Substituting this values in the equation of the circle give the locus of** $(h, k) \cdot p^2 + q^2 - p(p + x) - q(q + y) + q^2 = 0$

Q.13 Answer:

7

Solution:

Given, r^2 - $8r\sqrt{3}\cos\theta + \sin\theta + 15 = 0$

where $r\cos\theta = x$ and $y = r\sin\theta$

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$\therefore \quad \cos^2\theta + \sin^2\theta = \frac{x^2+y^2}{r^2} \Rightarrow r^2 = x^2 + y^2 \quad (\because \quad \cos^2\theta + \sin^2\theta = 1)$

Now it can be rewritten in Cartesian form as

$x^2 + y^2 - 8\sqrt{3}x + y + 15 = 0$

$\Rightarrow \quad x^2 + y^2 - 8\sqrt{3}x - 8y + 15 = 0$

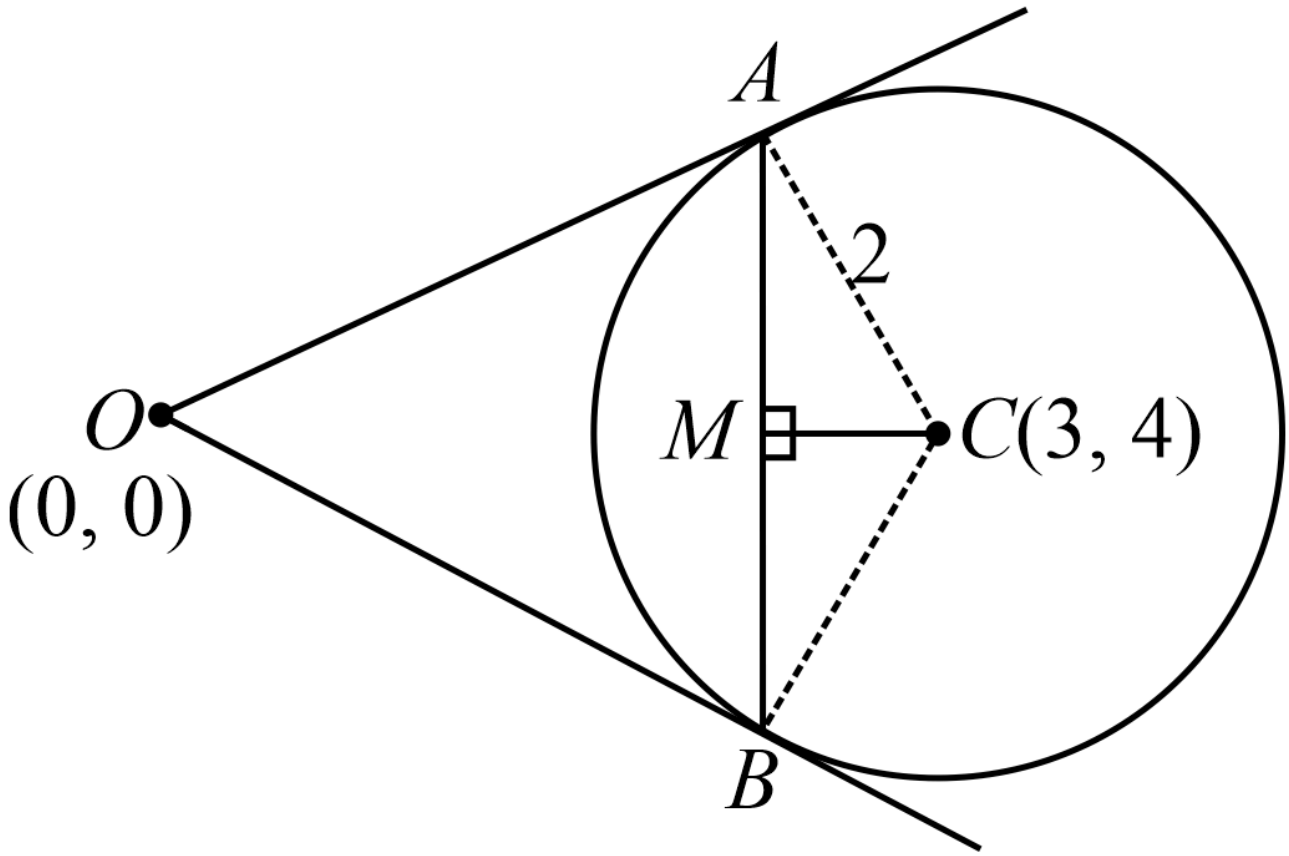
Now, radius $= \sqrt{4\sqrt{3}^2 + 4^2 - 15} = 7$

Q.14 Answer:

$\frac{4}{5}\sqrt{21}$

Solution:

Equation of chord of contact AB **is**



$3x + 4y - 21 = 0$

$\therefore CM = \frac{4}{5}$

$\therefore AM = \sqrt{4 - CM^2} = \frac{2}{5}\sqrt{21}$

$\therefore AB = \frac{4}{5}\sqrt{21}$

Q.15 Answer:

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a circle

Solution:

(b) Let $\mathrm{P(x, y)}$ be the point dividing the join of A and B in the ratio $2:3$ internally, then $x = \frac{20\cos\theta+15}{5} = 4\cos\theta + 3$

$\Rightarrow \cos\theta = \frac{x-3}{4}$

$\mathrm{y} = \frac{20\sin\theta+0}{5} = 4\sin\theta \Rightarrow \sin\theta = \frac{\mathrm{y}}{4}$

Squaring and adding (i) and (ii), we get the required locus $(x-3)^2 + y^2 = 16$, which is a circle

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